
DATA
METHOD
MODEL OUTPUT
1
Influence of environment on Danish avian communities
Atlas distributions of Danish breeding birds
Ecological characteristics
Migration
Species thermal index
Foraging guild
Environmental variables
Joint SDMs
Species distributions and influence of environment
1970s
1990s
2010s
Predicted richness
Variable importance through time
Significant variable effects across modelled species
Positive effect
Negative effect
2
Can we cluster distinct clusters of communities? Are the clusters consisten through time?
JSDM predictions
1. K-means clustering for regions of similar community composition
2. With 1970s clusters as a reference, which cluster are the grid cells in 1990 and 2010 most similar to?
Identification of grid cells with similar avian communities
Comparison of cluster ID for each grid cell at different moments in time
Identification of grid cells that 'flipped' cluster ID from 1970s to the 90s and 2010s
2.5
Environmental fingerprint of different clusters
1970s community clusters
JSDM environmental influence
Cluster-level environmental fingerprint - species environmental drivers weighed by species probability in said cluster
Environmental fingerprint of each cluster
3
> 35% of clusters 'flip' into the dominant cluster. What is underpinning 'flipping' from one cluster ID in 1970s to the dominant cluster in 2010s?
Grid cells that 'flipped' into dominant profile from 1970s
Ecological characteristics
Migration
Species thermal index
Foraging guild
Grid-level, species level changes in probability of occurrence 1970s to 2010s
Are certain groups of species more inclined to increase / decrease in grid cells that flipped from cluster 1 or 2 to cluster 3 (the 'dominant' cluster), from the 1970s to the 2010s?
- Across grid cells that flipped to the dominant cluster, per species, get the average gain/loss in predicted probability of occurrence
- Get the gains/losses of all species in each 'trait group'
- normalize to an 'average gain/loss of this trait group per species' so the results aren't skewed by different groupsizes
Migratory strategies
Thermal affinity groups
1970s cluster distribution
3.5
Can any changes of specific 'groups' of species gains/loss be explained through the lens of environmental variables
Joint SDMs
4
What is the spatial fingerprint of warm-affinity bird expansion? Is community warming driven purely by warm-affinity expansions or also by cold-affinity losses?
Ecological characteristics
Species thermal index
Predicted species distributions
Community weighted mean of species thermal index
Identification of grid cells dominated by cold vs warm affinity species
1990s minus 1970s
2010s minus 1970s
1990s (median -0.02)
2010s (median +0.02)
Dominant thermal group
very cold
cold
medium
warm
very warm